

Horizon singularity, energy conditions and shadows in time-dependent and spherically symmetric spacetime

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Horizon singularity, energy conditions and shadows in time-dependent and spherically symmetric spacetime

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Some dynamical spacetime solutions that describe ultra-compact objects are analysed. We show that by turning a static spacetime that describes a one-way wormhole into a dynamical time-dependent one, a singularity might arise in general. Several well-known wormhole solutions are studied, which are shown to become singular when promoted to a time-dependent wormhole. However, we show that one can construct a dynamical wormhole free of singularities. The corresponding Lagrangian that holds such dynamical spacetimes is also reconstructed, based on a two-scalar-field model free of ghosts. Energy conditions are also analysed, which might remain satisfied for a dynamical -singular- wormhole. Finally, the evolution of the optical appearance of such dynamical objects is simulated and compared to static solutions. Some interesting features emerge that point to a way to measure the time evolution of dynamical objects by looking at the photon ring structure around the object.

Outline

- 1 Motivation
- 2 Singularity problems
 - Examples
- 3 Shadows of non-stationary wormholes
 - Ray-tracing method
 - Dynamical Simpson-Visser spacetime
 - Dynamical Hayward spacetime
- 4 Final remarks

Motivation

The McVittie solution¹

$$ds^2 = - \left(\frac{1 - \frac{M}{2a(t)R}}{1 + \frac{M}{2a(t)R}} \right)^2 dt^2 + \left(1 + \frac{M}{2a(t)R} \right)^4 a^2(t) \{ dR^2 + R^2 d\Omega^2 \} \quad (1)$$

describes a massive object embedded in a FLRW Universe.

¹McVittie 1933.

²Modesto and Rattu 2025.

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- Recently, it has been shown² that there is a **naked curvature singularity at the horizon**.
- Is this exclusive of the McVittie solution?
- Can this be avoided? If so, how?

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Singularity problems

Indeed, this is a **rather universal phenomenon**, that is, the horizon embedding in a dynamical background becomes a naked singularity for a large class of models.

We consider the following form of the metric a the **dynamical** and **spherically symmetric spacetime**

$$ds^2 = -e^{2\nu(t,r)} dt^2 + e^{2\lambda(t,r)} dr^2 + e^{2\xi(t,r)} d\Omega^2, \quad (2)$$

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- At $e^{2\nu(t,r)} = 0$, we have the horizon(s).
- There, **curvatures are singular in general**.

The Ricci scalar:

$$R = 2e^{-2\nu} \left\{ \ddot{\lambda} + \left(\dot{\lambda} - \dot{\nu} \right) \dot{\lambda} \right\} + 4e^{-2\nu} \left\{ \ddot{\xi} + \left(\dot{\xi} - \dot{\nu} \right) \dot{\xi} \right\} + \dots$$

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We further require that:

- $e^{2\lambda+2\nu}$ is neither singular nor vanishes at the horizon.
- e^{ξ} is neither singular nor vanishes at the horizon.

Then, the metric functions can be calculated as

$$e^{\lambda(t,r)} = \int^t dt_1 e^{\nu(t_1,r)} \int^{t_1} dt_2 \Lambda(t_2, r) \quad (3)$$

$$e^{\xi(t,r)} = \int^t dt_1 e^{\nu(t_1,r)} \int^{t_1} dt_2 \Xi(t_2, r) e^{\nu(t_2,r)}. \quad (4)$$

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Note that, for $e^\nu = e^{\nu(r)}$, the areal radius reads

$$e^{\xi(t,r)} = \Xi_1(r) + e^{2\nu(r)} \Xi_2(r, t), \quad \Xi_2(r, t) \equiv \int^t dt_1 \int^{t_1} dt_2 \Xi(t_2, r). \quad (5)$$

Examples: Simpson-Visser black bounce

We consider **exotic compact objects**, such as traversable wormholes and black holes with a throat.

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We consider **exotic compact objects**, such as traversable wormholes and black holes with a throat.

- First, the time-dependent extension of the Simpson-Visser black bounce³, whose metric is given by

$$e^{2\nu} = e^{-2\lambda} = 1 - \frac{2M}{\sqrt{r^2 + r_{\text{throat}}^2}}, \quad r \in (-\infty, \infty). \quad (6)$$

Choosing the functions in (5) as $\Xi_1(r) = \sqrt{r^2 + r_{\text{throat}}^2}$ and $\Xi_2(t, r) = e^{-2\nu(r)}\Xi_{02}(t)$, we get the areal radius function

$$e^\xi = \sqrt{r^2 + r_{\text{throat}}^2} + \Xi_{02}(t). \quad (7)$$

For $r > 0$, this geometry presents one event horizon at $r = \sqrt{4 - r_{\text{throat}}^2}$ when $r_{\text{throat}} < 2M$.

³Simpson and Visser 2019.

Examples: Hayward-like wormhole

- Next, we consider the time-dependent extension of the Hayward geometry⁴:

$$e^{2\nu} = e^{-2\lambda} = 1 - \frac{2Mr^2}{|r|^3 + 2M\gamma^2}, \quad r \in (-\infty, \infty). \quad (8)$$

Similarly, choosing the functions $\Xi_1(r) = \sqrt{r^2 + r_{\text{throat}}^2}$ and $\Xi_2(t, r) = e^{-2\nu(r)}\Xi_{02}(t)$ in (5), the areal radius function reads

$$e^\xi = \sqrt{r^2 + r_{\text{throat}}^2} + \Xi_{02}(t). \quad (9)$$

For $r > 0$, this geometry presents two event horizons when $\gamma < 4\sqrt{3}M/9 \approx 0.770M$.

⁴Hayward 2006.

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- We give general prescriptions on how this can be avoided.
- As examples, we construct the dynamical extensions of the Simpson-Visser black bounce and a Hayward-like wormhole. In these particular cases, **only** e^ξ **depends on** t .

Shadows of non-stationary wormholes

In order to simulate their optical appearances, we need to solve the **null geodesic equations**. In the plane $\vartheta = \pi/2$:

$$\frac{d}{d\sigma} (e^{2\nu} k^t) + \frac{(k^\varphi)^2}{2} \frac{\partial}{\partial t} (e^{2\xi}) = 0, \quad \frac{d}{d\sigma} (e^{2\xi} k^\varphi) = 0, \quad (10)$$

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Using the **adiabatic approximation** $\implies \partial_t e^\xi \approx 0$. We shall treat $\Xi_{02}(t) \equiv \Xi$ as an additional parameter, so we get two conserved quantities:

$$e^{2\nu} k^t = E, \quad e^{2\xi} k^\varphi = L. \quad (11)$$

Null geodesics only depend on their **impact parameter** $b = L/E$ and the **effective potential** $V(r) = e^{2\nu-2\xi}$

$$\left(\frac{dr}{d\sigma}\right)^2 = \frac{1}{b^2} - V(r). \quad (12)$$

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The (possibly several) maxima of the effective potential $V(r)$ are called **photon spheres** r_{ps} , which define the so-called **critical impact parameters**

$$b_c = \frac{1}{\sqrt{V(r_{\text{ps}})}}. \quad (13)$$

Geodesics approaching one of those b_c will turn an arbitrarily large number of times around the compact object in unstable circular orbits.

Ray-tracing method

We know simulate the shadow of the previous examples using the so-called **ray-tracing method**⁵.

Assumptions and simplifications:

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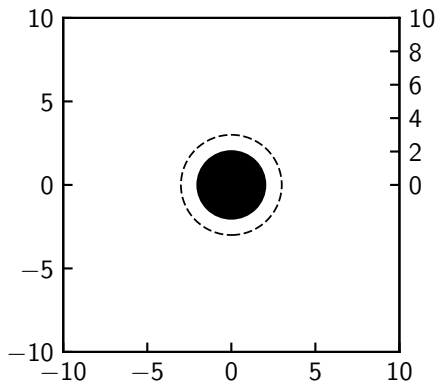
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- The emission is isotropic in the rest frame of the disk.

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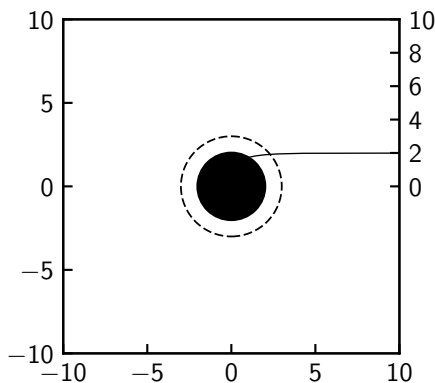
1. Compute the trajectory of the geodesic

$$\frac{d\varphi}{dr} = \pm \frac{b e^{-\xi}}{\sqrt{e^{2\xi} - b^2 e^{2\nu}}} . \quad (14)$$



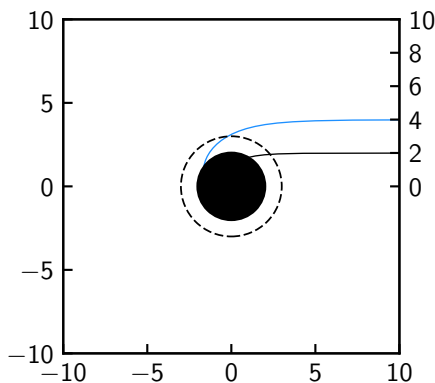
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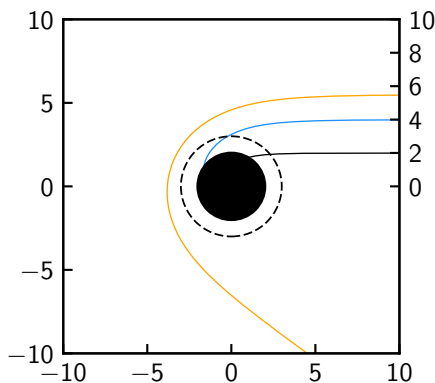
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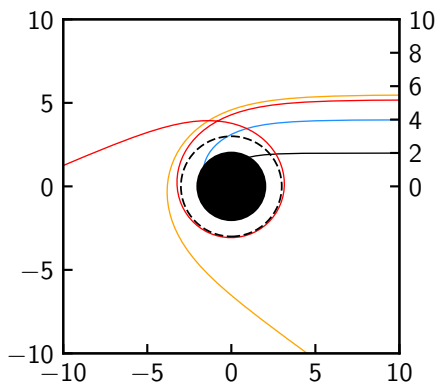
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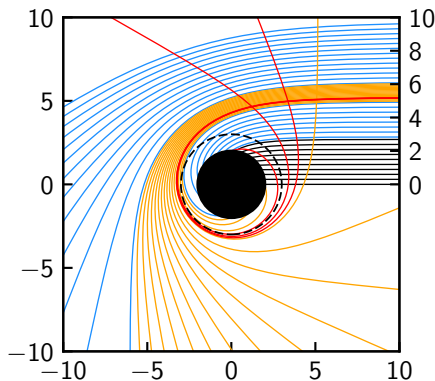
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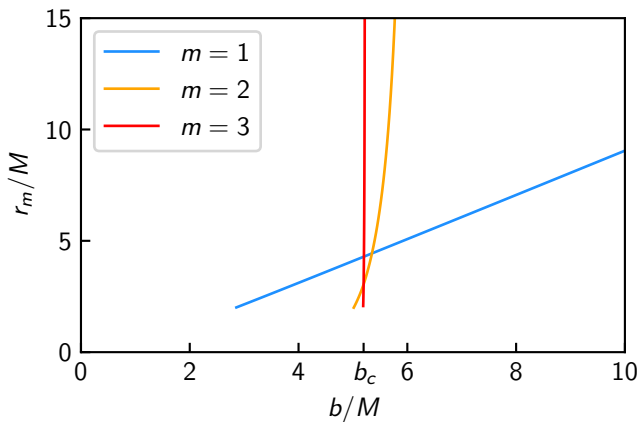


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2. A geodesic with impact parameter b , has its m -th intersection with the disk at some radial coordinate r . This information is stored in the **transfer functions** $r_m(b)$.



3. The disk emits with **specific intensity** I_{ω}^{em} in its rest frame. Since $I_{\omega}^{\text{em}}/\omega^3$ is conserved⁶ along any given geodesic

$$I_{\omega'}^{\text{ob}} = \left(\frac{\omega'}{\omega}\right)^3 I_{\omega}^{\text{em}} = e^{3\nu} I_{\omega}^{\text{em}} \implies I^{\text{ob}} = \int I_{\omega'}^{\text{ob}} d\omega' = e^{4\nu} I^{\text{em}}. \quad (15)$$

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4. The **observed intensity** profile is then the sum of all intersections with the disk

$$I^{\text{ob}}(b) = \sum_{m=1}^{\infty} e^{4\nu(r)} I^{\text{em}}(r) \Big|_{r=r_m(b)}. \quad (16)$$

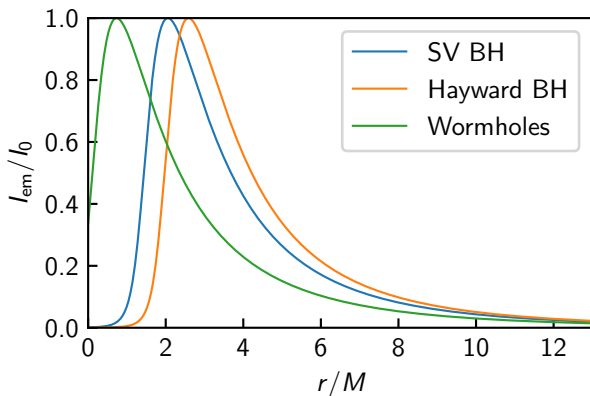
We truncate the sum at $m = 3$, since higher-order intersections contribute negligibly to the total intensity⁷.

⁶This may be derived from Liouville's theorem.

⁷Bisnovatyi-Kogan and Tsupko 2022; Gralla, Holz, and Wald 2019.

5. To model the emission, we use the parametrised family of toy-models⁸

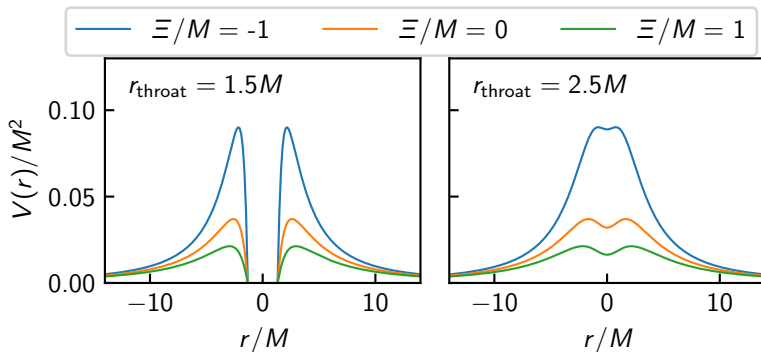
$$I^{\text{em}}(r; \mu, \sigma, \beta) = \frac{\exp\left(-\frac{1}{2}\left(\beta + \operatorname{arcsinh}\left(\frac{r-\mu}{\sigma}\right)\right)^2\right)}{\sqrt{(r-\mu)^2 + \sigma^2}}. \quad (17)$$

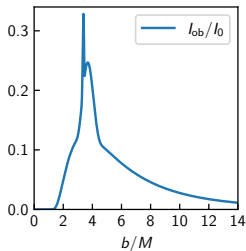
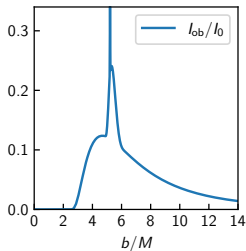
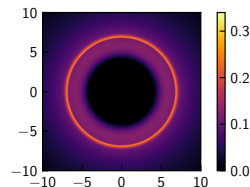
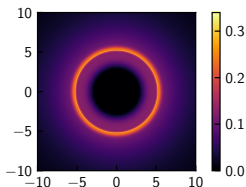
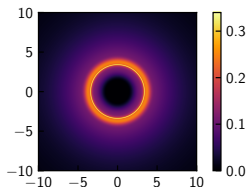
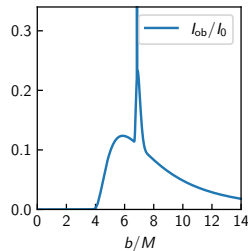


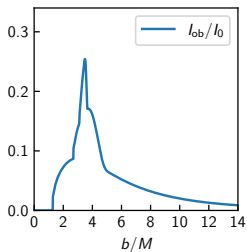
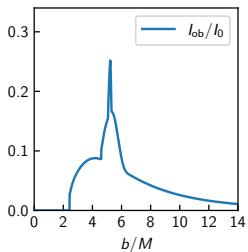
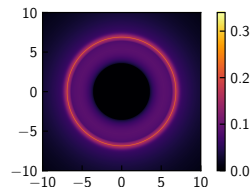
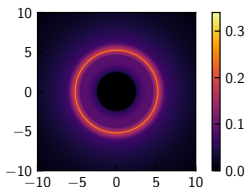
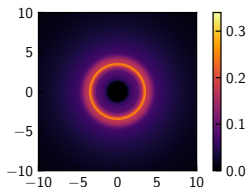
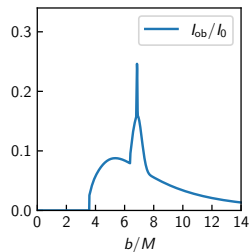
⁸Cárdenas-Avendaño, Lupsasca, and Zhu 2023; Gralla, Lupsasca, and

Dynamical Simpson-Visser spacetime

$$V(r) = \left(\sqrt{r^2 + r_{\text{throat}}^2} + \Xi \right)^{-2} \left(1 - \frac{2M}{\sqrt{r^2 + r_{\text{throat}}^2}} \right) \quad (18)$$

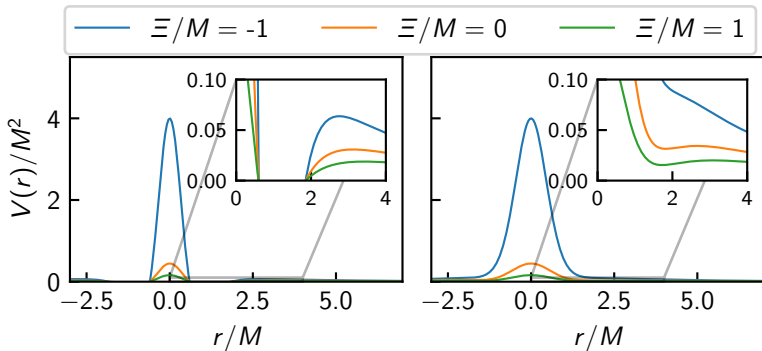


Simpson-Visser black hole ($r_{\text{throat}} = 1.5M$)(a) $\Xi/M = -1$ (b) $\Xi/M = 0$ (c) $\Xi/M = 1$ 

Simpson-Visser wormhole ($r_{\text{throat}} = 2.5M$)(a) $\Xi/M = -1$ (b) $\Xi/M = 0$ (c) $\Xi/M = 1$ 

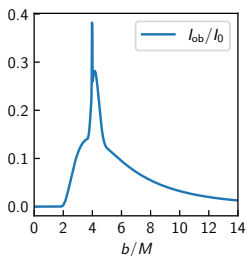
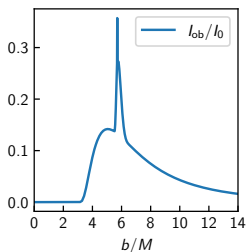
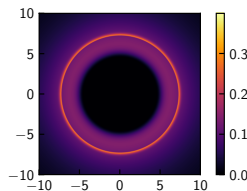
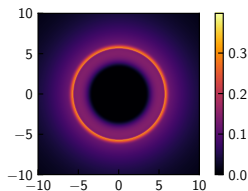
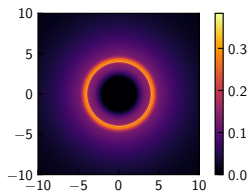
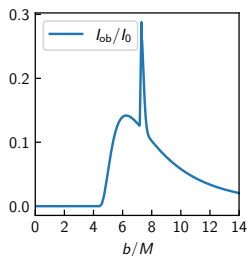
Dynamical Hayward spacetime

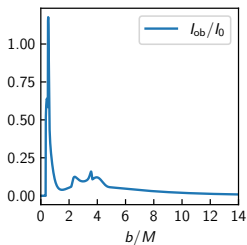
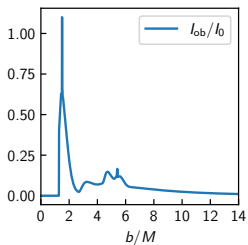
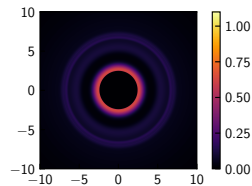
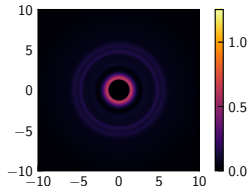
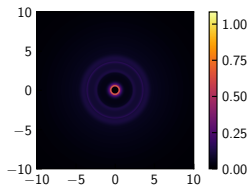
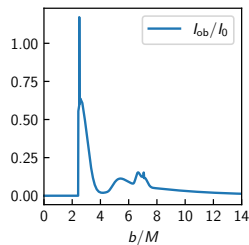
$$V(r) = \left(\sqrt{r^2 + r_{\text{throat}}^2} + \Xi \right)^{-2} \left(1 - \frac{2Mr^2}{|r|^3 + 2M\gamma^2} \right) \quad (19)$$



$\gamma = 0.5M$

$\gamma = 1.0M$

Hayward black hole ($\gamma = 0.5M$)(a) $\Xi/M = -1$ (b) $\Xi/M = 0$ (c) $\Xi/M = 1$ 

Hayward wormhole ($\gamma = 1.0M$)(a) $\Xi/M = -1$ (b) $\Xi/M = 0$ (c) $\Xi/M = 1$ 

Final remarks

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- We study the optical appearance of some of these extensions using an **adiabatic approximation**.
- The shadow size increases with Ξ , while higher order rings narrow.
- This combined effect has the final result of **demagnifying the characteristic light rings**.
- The source code can be found on my GitHub page:
<https://github.com/alvaroslzar/GRAVITYp>

Thank you for your attention

Questions?

Singularity problems

The Ricci scalar:

$$R = 2e^{-2\nu} \left\{ \ddot{\lambda} + (\dot{\lambda} - \dot{\nu}) \dot{\lambda} \right\} + 4e^{-2\nu} \left\{ \ddot{\xi} + (\dot{\xi} - \dot{\nu}) \dot{\xi} \right\} + \dots$$

For the Ricci scalar to be regular, we require

$$\ddot{\lambda} + (\dot{\lambda} - \dot{\nu}) \dot{\lambda} = \Lambda_0(t, r) e^{2\nu}, \quad (20)$$

$$\ddot{\xi} + (\dot{\xi} - \dot{\nu}) \dot{\xi} = \Xi_0(t, r) e^{2\nu}, \quad (21)$$

where $\Lambda_0(t, r)$ and $\Xi_0(t, r)$ are smooth.

Time-independent horizon

- The metric function e^λ can be calculated as

$$e^{\lambda(t,r)} = \int^t dt_1 e^{\nu(t_1,r)} \int^{t_1} dt_2 \Lambda(t_2, r) . \quad (22)$$

- e^λ is singular at $e^\nu = 0$, which is due to the integration “constants” w.r.t. t_1 and t_2 .
- Since those integration “constants” are only functions of r , the horizon is time-independent.



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